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PROJECT REPORT

AI-POWERED E-COMMERCE DEVOPS & CLOUD-NATIVE DELIVERY PLATFORM

Project Title	AI-POWERED E-COMMERCE DEVOPS & CLOUD-NATIVE DELIVERY PLATFORM
Internship Domain	DEVOPS
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Submission Date	9th October 2026

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1. Project Overview

1.1 Project Description

Nexvion – AI-Powered E-Commerce DevOps & Cloud-Native Delivery Platform is a DevOps and cloud-native project developed to automate and manage the complete software delivery lifecycle of an existing e-commerce web application.

The project focuses on the **deployment, automation, security, infrastructure, monitoring, logging, and operational management** of the application rather than its application-level development. The platform integrates GitHub and Jenkins for source control and CI/CD automation, Docker for containerization, Terraform and Ansible for infrastructure and server configuration, and Amazon EKS with Helm for Kubernetes-based deployment.

The CI/CD pipeline performs code validation, dependency checking, secret scanning using GitLeaks, Docker image creation, vulnerability scanning using Trivy, image publishing to Amazon ECR, and automated deployment to an Amazon EKS cluster. The deployed application is monitored using Prometheus and Grafana, while Fluent Bit, Elasticsearch, and Kibana provide centralized application logging.

An additional incident-analysis component was implemented in Python to collect information from Kubernetes, Elasticsearch, and Prometheus and generate structured diagnostic reports. Google Gemini is used for AI-assisted analysis when an API key is available, with a rule-based analysis mode available as a fallback.

The project also implements deployment health verification and automated Helm rollback handling so that a failed Kubernetes deployment can be detected and the previously working release can be restored.

1.2 Use Case

The primary use case of Nexvion is to provide a **single automated delivery workflow for deploying and operating a containerized web application on AWS**.

In the implemented workflow, changes are maintained in GitHub and processed through Jenkins. The pipeline validates the application, performs security checks, builds and scans the Docker image, and publishes the validated image to Amazon ECR. Jenkins then deploys the image to the Amazon EKS cluster using the Nexvion Helm chart.

After deployment, the application is verified through Kubernetes health checks and service-level checks. Runtime metrics are collected through Prometheus and visualized using Grafana, while application logs are collected by Fluent Bit and stored in Elasticsearch for analysis through Kibana.

If a deployment fails during the rollout process, Jenkins collects diagnostic information and invokes the incident-analysis tooling before performing a dynamic Helm rollback to the previously deployed release. This provides an integrated workflow from **source code to deployment, monitoring, troubleshooting, and recovery**.

The project is implemented as a **staging-oriented AWS environment**, using an EKS cluster with a single t3.small worker node. The deployment uses a capacity-safe rolling update strategy suitable for the available cluster resources.

2. Problem Statement

2.1 Existing Problem

The primary use case of Nexvion is to provide a **single automated delivery workflow for deploying and operating a containerized web application on AWS**.

In the implemented workflow, changes are maintained in GitHub and processed through Jenkins. The pipeline validates the application, performs security checks, builds and scans the Docker image, and publishes the validated image to Amazon ECR. Jenkins then deploys the image to the Amazon EKS cluster using the Nexvion Helm chart.

After deployment, the application is verified through Kubernetes health checks and service-level checks. Runtime metrics are collected through Prometheus and visualized using Grafana, while application logs are collected by Fluent Bit and stored in Elasticsearch for analysis through Kibana.

If a deployment fails during the rollout process, Jenkins collects diagnostic information and invokes the incident-analysis tooling before performing a dynamic Helm rollback to the previously deployed release. This provides an integrated workflow from **source code to deployment, monitoring, troubleshooting, and recovery**.

The project is implemented as a **staging-oriented AWS environment**, using an EKS cluster with a single t3.small worker node. The deployment uses a capacity-safe rolling update strategy suitable for the available cluster resources.

2.2 Proposed Approach

The project addresses this problem by implementing an integrated DevOps and cloud-native delivery platform around the existing Nexvion e-commerce workload.

GitHub is used for source-code management, while Jenkins automates the delivery pipeline. The pipeline performs validation and dependency checks, GitLeaks secret scanning, Docker image creation, Trivy vulnerability scanning, and publishing of validated images to Amazon ECR.

Infrastructure and configuration are managed using Terraform and Ansible, while Amazon EKS provides the Kubernetes environment for application deployment. Helm is used to package and manage the application release.

For operational management, Metrics Server and HPA provide resource-based scaling information, Prometheus and Grafana provide monitoring, and Fluent Bit with Elasticsearch and Kibana provide centralized logging. A Python-based incident-analysis tool combines Kubernetes, Elasticsearch, and Prometheus information and can use Google Gemini for AI-assisted analysis.

Deployment health checks and Helm-based rollback handling complete the workflow, allowing a failed release to be diagnosed and the previously deployed release to be restored.

In short, the proposed approach connects the complete lifecycle into one workflow:
Code → Validate → Secure → Build → Scan → Publish → Deploy → Verify → Monitor → Log → Analyse → Recover

3. Project Objectives

3.1 Primary Objectives

The primary objective of the project is to design and implement a **complete, automated, secure, observable, scalable, and repeatable DevOps and cloud-native delivery platform** around the existing Nexvion e-commerce application.

The project aims to integrate the complete software delivery lifecycle rather than demonstrating individual DevOps tools independently. The platform connects source control, automation, containerization, CI/CD, infrastructure provisioning, configuration management, Kubernetes deployment, monitoring, centralized logging, security practices, cloud deployment, and AI-assisted incident analysis.

3.2 Specific Objectives

The major objectives of the project are:

1. **Source Code and Collaboration**
   - Manage the project source code using Git and GitHub.
   - Establish a structured and version-controlled development workflow.

2. **Automation**
   - Automate repetitive operational and deployment activities using Linux, Bash, Jenkins, Terraform, and Ansible.

3. **Containerization**
   - Package the existing e-commerce workload into a secure and consistent Docker container.
   - Use Docker Compose for local and staging execution.

4. **CI/CD Implementation**
   - Build an automated Jenkins pipeline covering validation, security scanning, image building, registry publishing, deployment, and health verification.

5. **DevSecOps Integration**
   - Integrate secret detection and container vulnerability scanning into the delivery pipeline.
   - Prevent releases containing unacceptable security findings.

6. **Infrastructure Automation**
   - Provision and manage cloud infrastructure using Terraform.
   - Automate server configuration using Ansible.

7. **Kubernetes Deployment**
   - Deploy the application on Amazon EKS using Kubernetes and Helm.
   - Implement health probes, resource controls, rolling updates, services, ingress, and horizontal scaling.

8. **Monitoring and Observability**
   - Monitor infrastructure and application-related metrics using Prometheus and Grafana.
   - Provide operational visibility into deployment health and resource utilization.

9. **Centralized Logging**
   - Collect and centralize Kubernetes container logs using Fluent Bit, Elasticsearch, and Kibana.
   - Enable centralized investigation of application and platform events.

10. **AI-Assisted Incident Analysis**
    - Develop an incident-analysis workflow that combines Kubernetes, Elasticsearch, and Prometheus evidence.
    - Support AI-assisted analysis to help classify and understand operational incidents.

11. **Deployment Reliability**
    - Implement health verification and controlled rollback mechanisms.
    - Ensure that failed releases can be diagnosed and recovered using the previously deployed version.

12. **Cloud-Native Integration**
    - Integrate all major components into a connected cloud-based DevOps lifecycle rather than operating them as isolated demonstrations.

These objectives follow the project's intended final state: version-controlled application code, automated operations, containerized execution, Jenkins CI/CD, Terraform infrastructure, Ansible configuration, Kubernetes deployment, monitoring, centralized logging, integrated security, controlled releases, cloud deployment, and AI-assisted incident analysis.

4. Project Scope

The project covers the complete DevOps lifecycle surrounding the existing Nexvion e-commerce application. The application itself is treated as the workload, while the primary scope is the design and implementation of its automated, secure, cloud-native delivery and operational ecosystem. This is consistent with the project specification, which emphasizes integration of the technologies into one connected DevOps platform rather than isolated demonstrations.

4.1 Included Scope

The following functions, technologies, and activities are included in the project:

1. **Application Environment Analysis**
   The existing e-commerce application was analysed to understand its runtime requirements, dependencies, application ports, health endpoints, configuration requirements, and operational behaviour. This information was used to design the containerization and deployment environment.
2. **Source Control and Collaboration**
   Git and GitHub were used for version control, repository management, branching, commit history, documentation, and controlled source-code collaboration.
3. **Linux and Automation**
   Linux was used as the primary operational environment. Bash and shell-based automation were used for environment checks, application operations, health verification, cleanup, and other repetitive administrative tasks.
4. **Containerization**
   Docker was used to create a consistent and secure runtime for the application. Docker Compose was used for local and staging-oriented container execution. The container was hardened using a non-root user, restricted Linux capabilities, read-only filesystem configuration, temporary filesystems, and security headers.
5. **CI/CD Pipeline**
   Jenkins was integrated as the primary automation server. The pipeline covers source checkout, validation, dependency checking, secret scanning, Docker image creation, vulnerability scanning, Amazon ECR publishing, Kubernetes deployment, health verification, diagnostics, and deployment rollback.

6. **Infrastructure and Configuration Management**
   Terraform was used for Infrastructure as Code, including AWS infrastructure and EKS resources. Ansible was used for server-side configuration and installation of required operational components.
7. **Kubernetes and Helm Deployment**
   Amazon EKS was used as the cloud-native orchestration platform. Kubernetes resources include namespaces, deployments, services, ConfigMaps, Secrets, Ingress, probes, resource controls, and Horizontal Pod Autoscaling. Helm was used for repeatable application packaging and deployment.
8. **Monitoring and Observability**
   Metrics Server, Prometheus, and Grafana were integrated to provide visibility into Kubernetes nodes, application pods, deployment replicas, and HPA behaviour.
9. **Centralized Logging**
   Fluent Bit, Elasticsearch, and Kibana were implemented to collect, store, and investigate application/container logs centrally within the Kubernetes environment.
10. **DevSecOps Integration**
    Security was integrated into the delivery lifecycle through dependency scanning, GitLeaks secret detection, Trivy container image scanning, secure container configuration, least-privilege practices, and Kubernetes security controls.
11. **AI-Assisted Incident Analysis**
    A Python-based incident analysis system was implemented to collect information from Kubernetes, Elasticsearch, and Prometheus and generate incident classification, severity assessment, investigation guidance, and remediation suggestions. Gemini-assisted analysis is supported, with a rule-based fallback when an AI provider is unavailable.
12. **Cloud Deployment and Reliability**
    The completed platform was deployed on AWS using Amazon EKS and Amazon ECR. Automated health checks, controlled rolling updates, deployment diagnostics, and dynamic Helm rollback were implemented to improve deployment reliability.

The overall scope therefore covers the complete lifecycle:
Code → Validate → Secure → Build → Scan → Publish → Deploy → Verify → Monitor → Log → Analyse → Recover

4.2 Limitations

The current implementation is a **production-style staging platform** and has several intentional limitations:

1. The primary e-commerce application development is outside the main scope. The existing application is used as the workload for demonstrating DevOps and cloud-native practices.
2. The EKS environment uses a **single t3.small worker node** with limited cluster capacity. Therefore, the Kubernetes rolling-update configuration uses maxSurge: 0 and maxUnavailable: 1 as a capacity-safe strategy rather than claiming zero-downtime deployment.
3. The NGINX Ingress Controller is exposed through a **NodePort (31449)** for internal validation. Public Internet exposure and production DNS routing were not implemented or validated.
4. The project does not currently use an AWS Application Load Balancer or Network Load Balancer, Route 53 production DNS, or production TLS termination.
5. Elasticsearch is configured as a **single-node staging deployment** with ephemeral storage. It is therefore not designed for highly available or long-term production log retention.
6. Prometheus and Grafana are configured for the constrained staging environment and are not deployed as a highly available monitoring stack.

7. The HPA is configured for 2–5 replicas based on CPU utilization. The implemented test verified that metrics were available and the HPA was active, but the controlled load test did not demonstrate scaling beyond the minimum two replicas.
8. The AI incident-analysis module depends on an external AI provider when Gemini-assisted analysis is enabled. A deterministic rule-based fallback is provided when an AI API key is unavailable.
9. Production-grade secret management using services such as AWS Secrets Manager or an External Secrets solution is not part of the current staging implementation.
10. High-availability, multi-node/multi-AZ Kubernetes infrastructure, persistent Elasticsearch storage, production-grade ingress/load balancing, and advanced disaster-recovery mechanisms are considered future enhancements rather than current project requirements.

These limitations define the boundary of the implemented project while keeping the platform focused on demonstrating an integrated and repeatable DevOps lifecycle.

5. Technologies & Tools Used

The project uses a combination of DevOps, cloud-native, security, monitoring, logging, and AI technologies. The technologies were selected to support the complete software delivery lifecycle rather than to demonstrate individual tools independently. The project specification defines the overall technology stack and their intended roles.

5.1 Technology Stack

Category	Technology / Tool	Role in the Project
Operating System	**Linux**	Application, Jenkins, Docker, and server environment
Version Control	**Git**	Source-code and configuration version management
Collaboration	**GitHub**	Repository hosting, collaboration, documentation, and version history
Automation	**Bash / Shell Scripting**	Operational automation, health checks, cleanup, and system tasks
Web Server	**Nginx**	Web serving and application container runtime
Containerization	**Docker**	Application packaging and isolated runtime
Multi-Container Environment	**Docker Compose**	Local/staging container orchestration
CI/CD	**Jenkins**	Automated validation, security, build, publishing, deployment, and verification
Infrastructure as Code	**Terraform**	AWS and EKS infrastructure provisioning
Configuration Management	**Ansible**	Server configuration and environment setup
Container Orchestration	**Kubernetes / Amazon EKS**	Application deployment, scaling, networking, and workload management
Kubernetes Packaging	**Helm**	Repeatable Kubernetes application deployment and release management
Container Registry	**Amazon ECR**	Secure storage and distribution of Docker images
Kubernetes Networking	**NGINX Ingress Controller**	Internal application routing and HTTP access
Metrics Collection	**Metrics Server**	Kubernetes resource metrics and HPA functionality
Monitoring	**Prometheus**	Collection and querying of infrastructure and Kubernetes metrics

Visualization	**Grafana**	Monitoring dashboards and operational visibility
Centralized Logging	**Fluent Bit + Elasticsearch + Kibana (ELK)**	Log collection, storage, search, and investigation
Secret Scanning	**GitLeaks**	Detection of accidentally committed secrets
Dependency Security	**npm audit**	Dependency vulnerability checking
Container Security	**Trivy**	Docker image vulnerability scanning
AI-Assisted Analysis	**Python + Gemini AI**	Incident classification, severity analysis, investigation, and remediation assistance
Cloud Platform	**Amazon Web Services (AWS)**	Cloud infrastructure and deployment environment

5.2 Purpose of Each Tool

Git and GitHub provide the foundation for source control and collaboration. Project configuration, infrastructure code, Kubernetes manifests, Helm charts, documentation, and automation scripts are maintained under version control.

Docker and Docker Compose provide a consistent application runtime. Docker is also integrated into the Jenkins pipeline so that the same application image can progress through scanning, registry publishing, and deployment.

Jenkins acts as the central CI/CD automation engine. The pipeline connects validation, dependency checking, GitLeaks, Docker image creation, Trivy scanning, Amazon ECR publishing, EKS deployment, health verification, and rollback.

Terraform and Ansible address infrastructure and configuration automation. Terraform provisions the required AWS infrastructure, while Ansible is used for server configuration and operational setup.

Kubernetes, Amazon EKS, and Helm provide the cloud-native deployment layer. Kubernetes manages application replicas, services, configuration, health probes, scaling, and rolling updates, while Helm provides repeatable release management.

Prometheus, Grafana, and ELK provide the observability layer. Prometheus collects metrics, Grafana visualizes them, and Fluent Bit with Elasticsearch and Kibana provides centralized log collection and investigation.

GitLeaks and Trivy integrate security into the delivery lifecycle by detecting exposed secrets and vulnerabilities in container images. Dependency scanning is also included in the Jenkins validation stage.

Python and Gemini AI provide the intelligent incident-analysis layer. The analysis system combines Kubernetes, Elasticsearch, and Prometheus evidence to assist with incident classification, severity assessment, possible root-cause identification, investigation guidance, and remediation suggestions.

Overall, these technologies form a connected DevOps platform covering **source control, automation, containerization, CI/CD, infrastructure, orchestration, security, monitoring, logging, cloud deployment, and AI-assisted operations**.

6. System / Project Architecture

The architecture of the Nexvion DevOps and Cloud-Native Delivery Platform is organized into interconnected layers covering source control, CI/CD automation, infrastructure, container orchestration, security, observability, centralized logging, and AI-assisted incident analysis. The architecture is built around the existing e-commerce application and connects the individual DevOps technologies into a single delivery and operations platform.

6.1 Architecture Diagram

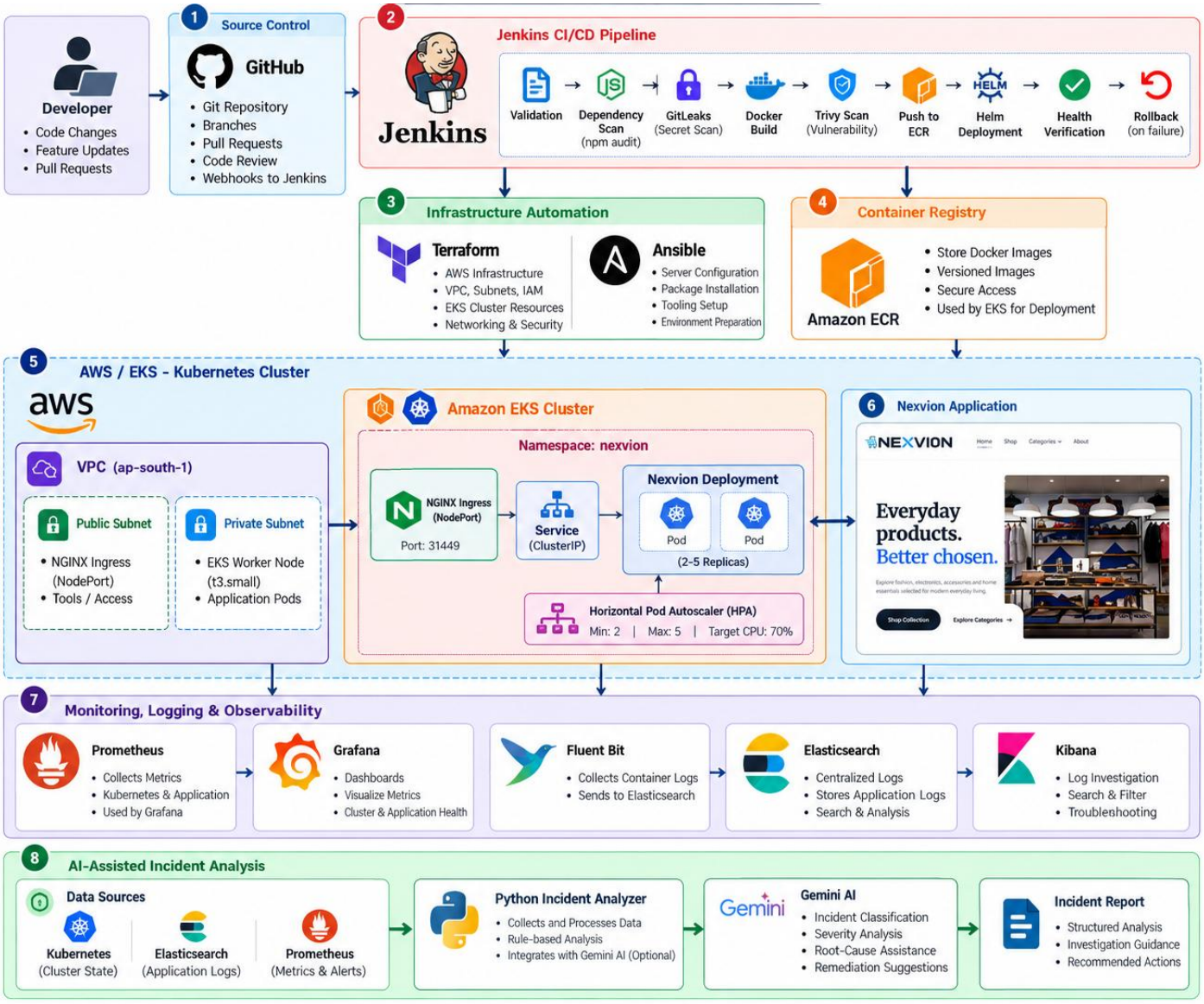


Figure: System Architecture of the Nexvion AI-Powered E-Commerce DevOps & Cloud-Native Delivery Platform

6.2 Component Description

1. **Source Control Layer**

   Git and GitHub form the source-control layer of the platform. The project repository contains the application workload together with Docker configuration, Jenkins pipeline definitions, Terraform infrastructure code, Ansible configuration, Kubernetes manifests, Helm charts, monitoring configuration, logging configuration, and AI incident-analysis components.

2. **CI/CD Automation Layer**

   Jenkins acts as the central automation component. It provides the controlled execution environment for the project's CI/CD pipeline. The pipeline integrates application validation, dependency checking, secret scanning, container image creation, vulnerability scanning, registry publishing, Kubernetes deployment, health verification, and rollback capabilities.

3. **Infrastructure Automation Layer**

   Terraform provides Infrastructure as Code for AWS resources and the Amazon EKS environment. **Ansible** complements Terraform by handling server configuration and operational environment setup.
   Together, these tools reduce dependence on manual infrastructure and server configuration.

4. **Container and Registry Layer**

   Docker packages the Nexvion application into a consistent container image. Docker Compose provides a convenient environment for local and staging-oriented execution.
   Amazon ECR acts as the container image registry. Images generated by the CI/CD pipeline are scanned and published to ECR before being used by the EKS deployment.

5. **Kubernetes Orchestration Layer**

   Amazon EKS provides the managed Kubernetes environment. The Nexvion workload is deployed in the nexvion namespace.
   The Kubernetes layer contains the application Deployment, Pods, Service, ConfigMap, Secret, Ingress configuration, resource constraints, health probes, and Horizontal Pod Autoscaler.
   Helm provides parameterized packaging and release management for the Nexvion application.

6. **Networking Layer**

   The Kubernetes **Service** provides internal connectivity to the application Pods. The **NGINX Ingress Controller** provides HTTP routing into the Kubernetes application environment.
   In the current staging implementation, the ingress controller uses a NodePort for internal validation rather than a production AWS load balancer.

7. **Monitoring Layer**

   Metrics Server provides Kubernetes resource metrics required for resource visibility and HPA operation.
   Prometheus collects and queries platform metrics, while **Grafana** presents the collected information through operational dashboards.
   The monitoring layer provides visibility into Kubernetes nodes, application Pods, deployment replicas, HPA state, and resource utilization.

8. **Centralized Logging Layer**

   Fluent Bit collects container logs from the Kubernetes environment and forwards them to **Elasticsearch**.
   Elasticsearch stores the collected logs, while **Kibana** provides a user interface for searching and investigating those logs.
   This creates a centralized logging layer instead of requiring administrators to inspect individual container logs manually.

9. **AI-Assisted Incident Analysis Layer**

   The **Python incident-analysis engine** combines operational evidence from Kubernetes, Elasticsearch, and Prometheus.

When configured with an AI provider, **Gemini** assists in interpreting the collected evidence. The system can provide incident classification, severity assessment, possible root-cause analysis, investigation guidance, and remediation suggestions.
A rule-based fallback is also available when AI-assisted analysis is not configured.

10. **Security Layer**
    Security controls are distributed throughout the architecture rather than implemented as a single isolated component.
    The platform incorporates:
    - GitLeaks for secret detection
    - npm audit for dependency checking
    - Trivy for container image vulnerability scanning
    - Non-root container execution
    - Restricted Linux capabilities
    - Read-only container filesystem
    - Kubernetes security contexts
    - Resource limits and requests
    - Least-privilege practices
    - Secure handling of environment secrets

    This architecture ensures that security controls are integrated into the software delivery and runtime environment.

7. Project Workflow

7.1 End-to-End Workflow

The Nexvion platform follows an integrated DevOps lifecycle in which source-code changes move through validation, security checks, containerization, registry publishing, Kubernetes deployment, verification, monitoring, logging, and incident analysis.

The overall workflow is:
Code → Validate → Secure → Build → Scan → Publish → Deploy → Verify → Monitor → Log → Analyse → Recover

This workflow connects the individual technologies into one continuous delivery process, which is the central principle of the project.

Step 1: Source Code and Collaboration
The workflow begins when a developer makes a change to the Nexvion application or its DevOps configuration.
The source code and project configuration are maintained in **Git and GitHub**, providing version history, branching, pull requests, and collaborative development.

Step 2: Jenkins Pipeline Trigger
The updated repository is retrieved by **Jenkins**. Jenkins acts as the central CI/CD automation engine and starts the defined pipeline.
The pipeline records the Git commit SHA so that the resulting container image and deployment can be associated with a specific version of the source code.

Step 3: Validation

The pipeline validates the project before building or deploying it.
The validation stage includes:

- Required project-file checks
- JavaScript syntax validation
- Docker Compose configuration validation
- Dependency/security validation
- Helm chart linting

If validation fails, subsequent deployment stages are stopped.

Step 4: Security Validation
Security checks are integrated into the pipeline.
GitLeaks scans the repository for accidentally exposed secrets, while dependency scanning checks application dependencies for known security issues.
This ensures that obvious security problems are detected before the application reaches the deployment stage.

Step 5: Docker Image Build
After successful validation and security checks, Jenkins builds the Nexvion Docker image.
The image is tagged using the Git commit SHA, allowing a specific source-code version to be associated with a specific container image.

Step 6: Container Vulnerability Scanning
The generated image is scanned using **Trivy**.
The pipeline evaluates the image for high and critical vulnerabilities. Images failing the configured security threshold are prevented from progressing through the delivery pipeline.

Step 7: Amazon ECR Publishing
After the image passes the security checks, Jenkins authenticates with **Amazon ECR** and publishes the image.
ECR provides the central container registry from which the EKS cluster retrieves the application image.

Step 8: Kubernetes Deployment
Jenkins configures access to the **Amazon EKS** cluster and validates the Helm chart before deployment.
The Nexvion application is then deployed using **Helm**, with the image repository and Git SHA supplied as deployment parameters.
The application is deployed into the nexvion Kubernetes namespace.

Step 9: Deployment and Health Verification
After the Helm deployment, Jenkins waits for the Kubernetes Deployment rollout to complete.
The pipeline verifies:

- Deployment availability
- Pod status
- Kubernetes Service
- Application health endpoint
- Application root page
- Products page
- Payment page

Health checks are performed at multiple levels to confirm that the deployed workload is functioning correctly.

Step 10: Controlled Failure and Rollback
If the deployment fails, Jenkins collects diagnostic information such as:

- Pod status
- Deployment description
- Kubernetes events
- Container logs
- Incident-analysis output

The pipeline identifies the previously deployed Helm revision and can perform a **dynamic Helm rollback** when required. After rollback, the application health endpoints are checked again to confirm recovery.

Step 11: Monitoring and Scaling

Once deployed, the application is monitored using **Metrics Server, Prometheus, and Grafana**.
Prometheus collects Kubernetes and workload metrics, while Grafana provides dashboards for operational visibility.
The Kubernetes **Horizontal Pod Autoscaler (HPA)** monitors CPU utilization and is configured with a minimum of 2 and maximum of 5 replicas.

Step 12: Centralized Logging

Application and container logs are collected by **Fluent Bit**.
The logs are forwarded to **Elasticsearch**, where they can be stored and searched. **Kibana** provides the interface for log investigation and troubleshooting.

Step 13: AI-Assisted Incident Analysis

When an operational problem occurs, the incident-analysis system can collect evidence from:

- Kubernetes
- Elasticsearch
- Prometheus

The Python-based incident analyzer processes the collected information and can use **Gemini AI** to assist with:

- Incident classification
- Severity assessment
- Possible root-cause identification
- Investigation guidance
- Remediation suggestions

A rule-based analysis mode is available as a fallback.

Step 14: Continuous Improvement

The resulting monitoring, logs, deployment information, and incident analysis provide operational feedback that can be used to improve future deployments and troubleshooting procedures.

8. Project Implementation

The implementation of the Nexvion DevOps and Cloud-Native Delivery Platform was carried out incrementally, beginning with application containerization and progressing through CI/CD, infrastructure automation, Kubernetes deployment, observability, centralized logging, AI-assisted incident analysis, and final end-to-end validation. The implementation focused on building the delivery ecosystem around the existing e-commerce workload rather than modifying the application's core business functionality.

8.1 Environment & Initial Setup

The implementation was performed using a Linux-based DevOps environment with Docker, Jenkins, AWS CLI, Kubernetes tools, Helm, Terraform, and Ansible.

The initial environment was prepared for containerized application execution and CI/CD automation. The existing Nexvion application was first analysed to identify its runtime requirements, application ports, health endpoint, dependencies, and web-serving requirements.

The major initial setup activities included:

- Preparing the Linux-based operational environment.
- Configuring Git and GitHub repository management.
- Installing and configuring Docker and Docker Compose.
- Creating the production-oriented Dockerfile and NGINX configuration.
- Configuring the /healthz health endpoint.
- Establishing Jenkins as the CI/CD automation server.
- Configuring required Jenkins pipeline tools and permissions.
- Preparing Terraform for AWS infrastructure provisioning.
- Preparing Ansible for server configuration.
- Configuring AWS CLI access for ECR and EKS operations.
- Preparing Kubernetes and Helm tooling.
- Creating the required AWS networking and EKS infrastructure.
- Creating the Amazon ECR repository for application images.

The application was initially validated locally using Docker Compose before progressing to cloud deployment.

8.2 Module / Feature Implementation

8.2.1 Containerization and Application Hardening

A Docker-based runtime was implemented for the Nexvion application using NGINX.
The container was hardened using:

- Non-root execution
- Restricted Linux capabilities
- Read-only root filesystem
- Temporary filesystems for required writable locations
- Resource controls
- Health checks
- HTTP security headers

Docker Compose was also configured to provide a repeatable local/staging execution environment.

8.2.2 CI/CD Pipeline

Jenkins was implemented as the central CI/CD platform.
The pipeline performs the following major activities:

1. Source-code checkout
2. Git commit identification
3. Project validation
4. Dependency scanning
5. GitLeaks secret scanning
6. Docker image build
7. Trivy vulnerability scanning
8. Amazon ECR authentication and image publishing
9. Helm validation
10. EKS deployment
11. Kubernetes rollout monitoring
12. Application health verification
13. Failure diagnostics
14. Dynamic Helm rollback when required

This connects the individual DevOps stages into one automated delivery process.

8.2.3 Infrastructure Automation

Terraform was implemented to manage the AWS infrastructure required by the project, including networking and EKS-related resources.

Ansible was used for server-side configuration and operational environment preparation.

This separation provides:

- **Terraform:** infrastructure provisioning
- **Ansible:** machine and environment configuration

8.2.4 Kubernetes and Helm

The application was migrated from container-based execution into an Amazon EKS Kubernetes environment.

Kubernetes resources implemented include:

- Namespace
- Deployment
- Pods
- Service
- ConfigMap
- Secret
- Ingress
- Resource requests and limits
- Liveness and readiness probes
- Horizontal Pod Autoscaler

The application was packaged using Helm so that deployment parameters such as the container image repository, image tag, replica count, resource limits, ingress configuration, and HPA settings could be managed consistently.

The current staging deployment uses a capacity-safe rolling-update configuration of:

- maxSurge: 0
- maxUnavailable: 1

This configuration was selected for the single-worker-node staging environment.

8.2.5 Security Implementation

Security controls were integrated throughout the implementation.

GitLeaks was configured for repository secret detection, while Trivy was integrated into the Jenkins pipeline for container vulnerability scanning.

The container and Kubernetes workloads also implement security controls including non-root execution, dropped capabilities, read-only filesystem configuration, restricted privilege escalation, and resource limits.

8.2.6 Monitoring and Observability

Metrics Server was installed to provide Kubernetes resource metrics and enable HPA operation.
Prometheus was implemented for metrics collection and querying, while Grafana was configured with a dedicated Nexvion observability dashboard.
The dashboard provides visibility into:

- Kubernetes node readiness
- Application replicas
- HPA state
- Pod resource usage
- Node CPU and memory usage
- Pod restarts
- Application workload health

8.2.7 Centralized Logging

A lightweight centralized logging stack was implemented using:
Fluent Bit → Elasticsearch → Kibana
Fluent Bit collects Kubernetes container logs and forwards them to Elasticsearch. Elasticsearch stores the logs, while Kibana provides search and investigation capabilities.
The implementation was validated by generating a unique test log entry and locating that entry in the Elasticsearch log index.

8.2.8 AI-Assisted Incident Analysis

A Python-based incident-analysis module was implemented to combine operational information from:

- Kubernetes
- Elasticsearch
- Prometheus

The analysis engine classifies incidents, evaluates severity, and provides investigation and remediation guidance.
Gemini AI integration was added as an optional AI-assisted analysis layer. A rule-based fallback remains available when an AI provider is not configured.
The API credential is maintained outside the source repository through environment configuration and is excluded from Git tracking.

8.3 Deployment / Execution

The final application deployment was performed on **Amazon Web Services using Amazon EKS**.

The deployment process consisted of the following major steps:

1. Terraform was used to provision and manage the required AWS infrastructure.
2. An Amazon ECR repository was prepared for the Nexvion container image.
3. Jenkins built the application image and assigned a Git SHA-based tag.
4. GitLeaks and Trivy security checks were executed.
5. The validated image was pushed to Amazon ECR.
6. Jenkins configured Kubernetes access to the nexvion-eks cluster.
7. Helm validated and deployed the Nexvion application.
8. Kubernetes created the application Pods and Service.
9. NGINX Ingress provided internal HTTP routing through the NodePort configuration.

10. Metrics Server and HPA provided resource-based scaling capability.
11. Prometheus and Grafana provided monitoring.
12. Fluent Bit, Elasticsearch, and Kibana provided centralized logging.
13. The AI incident analyzer provided operational analysis.
14. Jenkins performed post-deployment health verification.

The final validation demonstrated successful deployment with **2/2 Nexvion Pods ready**, successful application health checks, active HPA metrics, working Prometheus queries, functioning Grafana monitoring, centralized Elasticsearch logging, Kibana health, and successful AI-assisted incident analysis.

A controlled deployment failure was also tested. The pipeline detected the failed rollout, collected diagnostic information, identified the previous Helm release, performed a rollback, and subsequently verified application recovery. This demonstrated that deployment failure handling was integrated into the CI/CD process rather than being dependent solely on manual intervention.

9. Key Features / Functionalities

9.1 Feature List

The completed Nexvion DevOps and Cloud-Native Delivery Platform provides the following major system-level features:

1. Version-Controlled Development

Git and GitHub provide centralized source-code management, version history, branching, collaboration, and repository documentation.

2. Automated CI/CD Pipeline

Jenkins automates the complete software delivery process, including source checkout, validation, security checks, Docker image creation, registry publishing, Kubernetes deployment, and post-deployment verification.

3. Containerized Application

The Nexvion application runs as a Docker container using a hardened NGINX-based runtime, providing consistency between development, staging, and Kubernetes environments.

4. Automated Dependency and Secret Scanning

The pipeline performs dependency validation and GitLeaks scanning to identify potential security issues and accidentally exposed secrets before deployment.

5. Container Vulnerability Scanning

Trivy scans the generated Docker image for vulnerabilities. The pipeline enforces the configured HIGH and CRITICAL vulnerability threshold before allowing the image to proceed.

6. Amazon ECR Image Management

Docker images are tagged using the source-code Git SHA and published to Amazon ECR, providing traceability between source code, container images, and Kubernetes deployments.

7. Infrastructure as Code

Terraform provides repeatable AWS infrastructure provisioning, reducing dependency on manual cloud configuration.

8. Automated Server Configuration

Ansible automates server configuration and installation of required operational components, supporting a consistent Linux-based environment.

9. Kubernetes-Based Application Deployment
Amazon EKS manages the containerized Nexvion workload using Kubernetes Deployments, Pods, Services, ConfigMaps, Secrets, Ingress, health probes, and resource controls.

10. Helm-Based Release Management
Helm packages the Kubernetes application configuration and provides parameterized, repeatable deployment and release management.

11. Health Monitoring and Self-Recovery
Liveness and readiness probes continuously verify application health. Kubernetes can restart unhealthy containers, while Jenkins performs additional post-deployment application endpoint verification.

12. Horizontal Pod Autoscaling
The Kubernetes HPA monitors CPU utilization and is configured to maintain between 2 and 5 application replicas according to the configured utilization target.

13. Controlled Rolling Deployment
The application uses a capacity-safe rolling-update strategy appropriate for the single-node staging environment. Deployment failures can trigger diagnostic collection and Helm rollback to a previously deployed release.

14. Infrastructure and Application Monitoring
Metrics Server and Prometheus collect Kubernetes and workload metrics, while Grafana provides dashboards for operational visibility.

15. Centralized Logging
Fluent Bit collects container logs and forwards them to Elasticsearch, while Kibana provides centralized log search and investigation.

16. AI-Assisted Incident Analysis
The Python incident analyzer combines Kubernetes, Elasticsearch, and Prometheus information to classify incidents, assess severity, assist with root-cause investigation, and suggest remediation. Gemini AI can be used as the AI-assisted analysis provider.

17. Multi-Level Deployment Verification
The platform verifies deployments at multiple levels, including Pod-local application endpoints, Kubernetes Service connectivity, and internal NGINX Ingress NodePort access.

18. Failure Diagnostics and Rollback
When a deployment fails, Jenkins collects Kubernetes diagnostics, executes incident analysis, identifies the previous Helm release, and can roll back to the previous working deployment.

19. Security-Hardened Runtime
The application container and Kubernetes workload use security controls such as non-root execution, restricted capabilities, read-only filesystem configuration, resource limits, and restricted privilege escalation.

20. End-to-End Traceability
A major feature of the platform is traceability across the complete delivery chain:
Git Commit → Docker Image → ECR Image → Helm Release → Kubernetes Pod → Monitoring → Logs → Incident Analysis
This allows a deployed workload to be associated with the specific source-code version from which it was created.

10. Testing & Validation

The project was tested progressively from individual components to the complete integrated DevOps platform. Testing covered application validation, containerization, security scanning, CI/CD, cloud infrastructure, Kubernetes deployment, health verification, monitoring, centralized logging, AI-assisted incident analysis, and deployment recovery.

The final validation confirmed that the major components operate together as a connected delivery platform.

10.1 Testing Approach

Testing was performed at multiple levels:

1. Application-Level Testing

The application was checked for valid JavaScript syntax, correct Docker Compose configuration, successful container startup, and availability of important HTTP endpoints.

2. Container Testing

The Docker image was built successfully and checked for runtime health. The container configuration was also evaluated for security and operational correctness.

3. Security Testing

GitLeaks was used to detect exposed secrets, dependency scanning was performed, and Trivy was used to scan the Docker image for vulnerabilities.

4. CI/CD Testing

The Jenkins pipeline was executed to verify source checkout, validation, security scanning, Docker image creation, ECR publishing, Helm deployment, health verification, and failure-handling behavior.

5. Kubernetes Testing

The EKS deployment was tested for Pod readiness, Service connectivity, Ingress routing, resource metrics, HPA status, and Helm release state.

6. Observability Testing

Prometheus queries and Grafana dashboards were checked for live Kubernetes and application metrics.

7. Logging Testing

Application/container logs were generated and verified in Elasticsearch through the centralized Fluent Bit logging pipeline and Kibana.

8. AI Incident Analysis Testing

The incident analyzer was executed using Kubernetes, Elasticsearch, and Prometheus evidence, with Gemini-assisted analysis successfully demonstrated.

9. Failure and Recovery Testing

A controlled deployment failure was introduced to verify that Jenkins could detect the failed rollout, collect diagnostics, identify the previous Helm revision, perform rollback, and verify application recovery.

10.2 Test Cases

Test Case	Expected Result	Actual Result	Status
Git repository checkout	Repository successfully retrieved	Jenkins successfully checked out source	**PASS**

JavaScript syntax validation	No syntax errors	Validation completed successfully	**PASS**
Docker Compose validation	Valid Compose configuration	Configuration validated successfully	**PASS**
Dependency scan	No high/critical dependency issue	Validation passed	**PASS**
GitLeaks scan	No exposed secrets	0 leaks found	**PASS**
Docker image build	Image generated successfully	Git SHA-tagged image built	**PASS**
Trivy image scan	No HIGH/CRITICAL vulnerabilities	0 HIGH / 0 CRITICAL	**PASS**
ECR image publishing	Image available in registry	Image verified in Amazon ECR	**PASS**
EKS connectivity	Cluster accessible	EKS cluster accessible	**PASS**
Helm validation	Chart passes validation	Helm deployment completed successfully	**PASS**
Kubernetes rollout	Application Pods become ready	2/2 Pods ready	**PASS**
Pod-local health check	Application endpoints return HTTP 200	/healthz, /, /products.html, /payment.html returned 200	**PASS**
Service-level health check	Kubernetes Service routes correctly	HTTP 200 responses received	**PASS**
Ingress NodePort test	NGINX Ingress routes traffic	NodePort 31449 returned HTTP 200	**PASS**
HPA / Metrics Server	Metrics available and HPA active	CPU 1%/70%, 2–5 replicas configured	**PASS**
Prometheus	Live metrics available	Deployment metric returned successfully	**PASS**
Grafana	Dashboard displays live metrics	Nexvion observability dashboard verified	**PASS**
Centralized logging	Application logs reach Elasticsearch	nexvion-logs-* index verified	**PASS**
Kibana	Logs searchable	Kibana health green and logs searchable	**PASS**
AI incident analysis	Incident report generated	Gemini-assisted analysis executed	**PASS**
Deployment rollback	Failed release can be recovered	Helm rollback successfully restored previous release	**PASS**
Git hygiene	Secrets not committed	.env ignored and no secrets committed	**PASS**

10.3 Validation

The final system was validated through an end-to-end execution rather than only through isolated component testing.

The final validation confirmed:

- **GitLeaks:** 0 detected leaks.
- **Trivy:** 0 HIGH and 0 CRITICAL vulnerabilities.
- **Amazon ECR:** Git SHA-tagged image successfully published and verified.
- **Amazon EKS:** Nexvion deployment successfully rolled out with 2/2 Pods ready.
- **Helm:** Release successfully deployed and rollback behavior verified.
- **Application:** Four important endpoints returned HTTP 200.
- **Ingress:** Internal NGINX NodePort routing successfully returned HTTP 200.
- **HPA:** Metrics Server active and HPA operating with a 2–5 replica configuration.
- **Prometheus:** Live Kubernetes metrics successfully queried.
- **Grafana:** Nexvion observability dashboard successfully displayed live metrics.
- **ELK:** Application logs successfully reached Elasticsearch and were available for investigation.
- **Kibana:** Logging interface reported healthy status.

- **AI Analysis:** Gemini-assisted incident analysis successfully generated an incident report.
- **Rollback:** Controlled deployment failure was detected and successfully recovered through Helm rollback.
- **Repository Security:** .env remained ignored and no secret credentials were committed.

The final end-to-end validation was therefore marked **PASSED**, demonstrating that the implemented platform successfully integrates CI/CD, security, containerization, ECR, EKS, Helm, health verification, observability, centralized logging, AI-assisted analysis, and deployment recovery.

11. Results & Outcomes

The completed project successfully established an integrated DevOps and cloud-native delivery platform around the Nexvion e-commerce workload. The final implementation was validated across CI/CD, security, containerization, AWS infrastructure, Kubernetes, monitoring, centralized logging, AI-assisted analysis, and deployment recovery.

11.1 Final Results

The major implementation results are summarized below:

1. Automated Software Delivery

A Jenkins-based CI/CD pipeline was successfully implemented to automate the major stages of software delivery, from source-code checkout through application deployment and health verification.

2. Secure Container Delivery

The Nexvion application was successfully containerized using Docker. GitLeaks and Trivy were integrated into the delivery pipeline.

Final validation recorded:

- **GitLeaks:** 0 leaks detected
- **Trivy:** 0 HIGH/CRITICAL vulnerabilities
- Docker image successfully built and tagged using the Git SHA

3. Container Registry Integration

Amazon ECR was successfully integrated as the container registry.

The validated application image was published and verified in ECR using a Git SHA-based tag, providing traceability between the source revision and deployed image.

4. Cloud-Native Kubernetes Deployment

The application was successfully deployed to Amazon EKS using Helm.

The final environment demonstrated:

- EKS cluster operational
- Nexvion namespace active
- 2/2 application Pods ready
- ClusterIP Service operational
- NGINX Ingress NodePort operational
- Helm release successfully deployed

The final validated deployment used a capacity-safe rolling strategy of maxSurge: 0 and maxUnavailable: 1 for the single-worker staging environment.

5. Application Health Verification

The deployed application was successfully verified at multiple levels.

The following endpoints returned HTTP 200:

- /healthz
- /
- /products.html
- /payment.html

Service-level and internal Ingress NodePort testing also returned successful HTTP responses.

6. Monitoring and Observability

Metrics Server, Prometheus, and Grafana were successfully integrated.

The platform successfully provided:

- Kubernetes resource metrics
- Deployment replica information
- HPA metrics
- Node resource information
- Application Pod resource information
- Grafana visualization

The HPA was active with a configured range of **2–5 replicas** and a CPU target of **70%**.

7. Centralized Logging

The Fluent Bit → Elasticsearch → Kibana pipeline successfully collected and stored Nexvion application/container logs.

The nexvion-logs-* indices were verified and Kibana reported a healthy state, allowing centralized log search and investigation.

8. AI-Assisted Incident Analysis

The Python incident-analysis system successfully collected operational evidence from Kubernetes, Elasticsearch, and Prometheus.

Gemini-assisted analysis was successfully executed to provide incident classification, severity analysis, investigation guidance, and remediation-oriented recommendations.

9. Deployment Failure Recovery

A controlled deployment failure was successfully tested.

The system demonstrated the ability to:

1. Detect the failed Kubernetes rollout.
2. Collect deployment and Pod diagnostics.
3. Identify the previously deployed Helm revision.
4. Perform a Helm rollback.
5. Wait for the restored deployment.
6. Re-run application health checks.
7. Confirm successful recovery.

This demonstrated that deployment reliability and recovery were incorporated into the CI/CD lifecycle rather than relying entirely on manual recovery.

11.2 Outcome Evaluation

The project achieved the following overall outcomes:

Improved Automation

Manual activities across application validation, image building, security scanning, registry publishing, deployment, and health verification were converted into an integrated Jenkins workflow.

Improved Security

Security checks were incorporated directly into the delivery lifecycle through GitLeaks, dependency validation, Trivy image scanning, and hardened container/Kubernetes configurations.

Improved Deployment Consistency
Docker, Terraform, Ansible, Kubernetes, and Helm provide repeatable mechanisms for application execution, infrastructure provisioning, environment configuration, and deployment.

Improved Operational Visibility
Prometheus and Grafana provide metrics-based visibility, while Fluent Bit, Elasticsearch, and Kibana provide centralized logging and investigation capabilities.

Improved Incident Response
The incident-analysis system combines Kubernetes state, metrics, and logs to assist with understanding operational failures. Gemini integration further extends this capability through AI-assisted analysis.

Improved Deployment Reliability
Automated health checks, controlled rolling updates, deployment diagnostics, and Helm rollback provide a structured mechanism for detecting and recovering from deployment failures.

End-to-End Traceability
The platform establishes traceability across the delivery lifecycle:

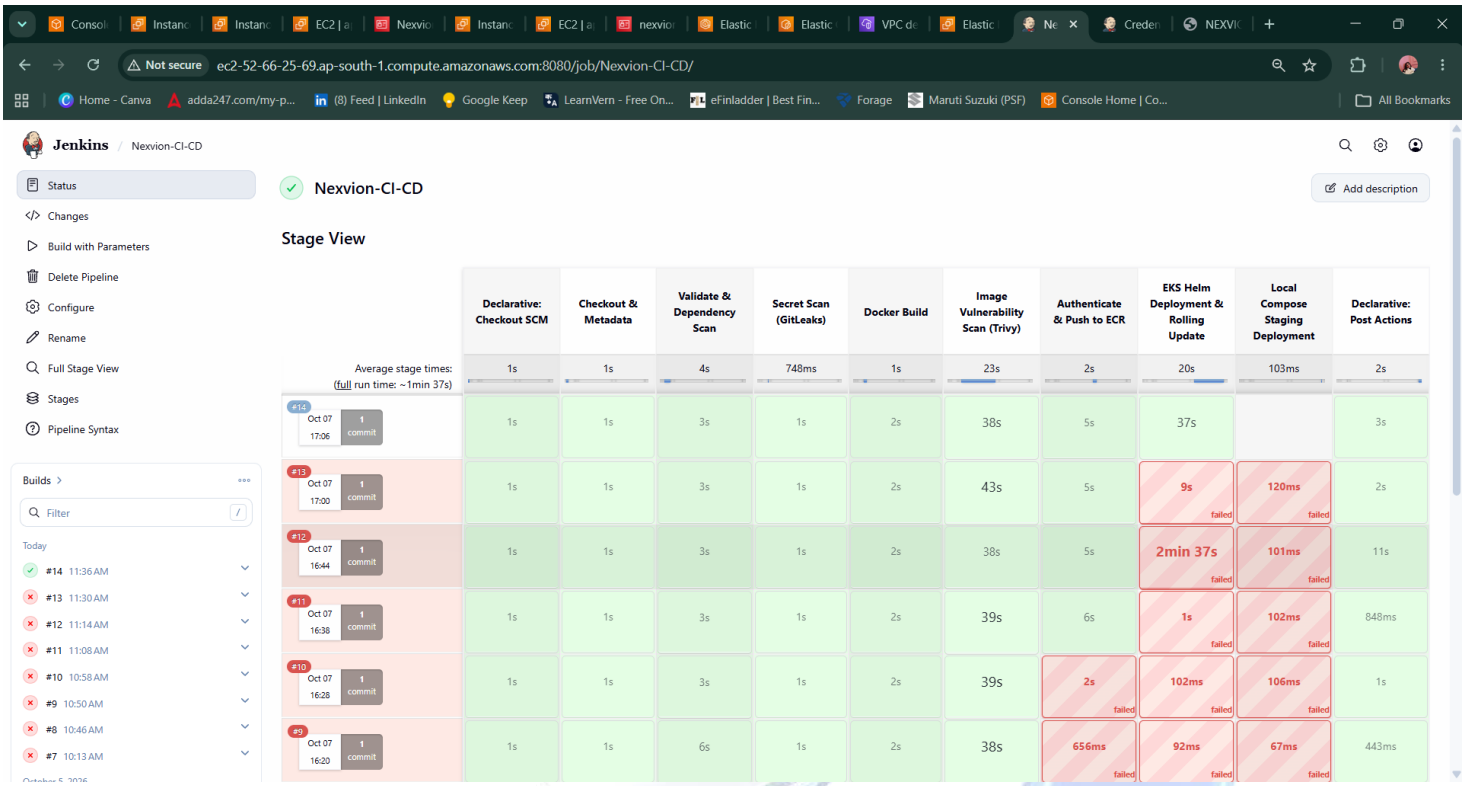
Git Commit → Docker Image → ECR → Helm Release → Kubernetes Pod → Metrics → Logs → Incident Analysis
This provides a clear relationship between the source version, deployed artifact, runtime environment, and operational evidence.

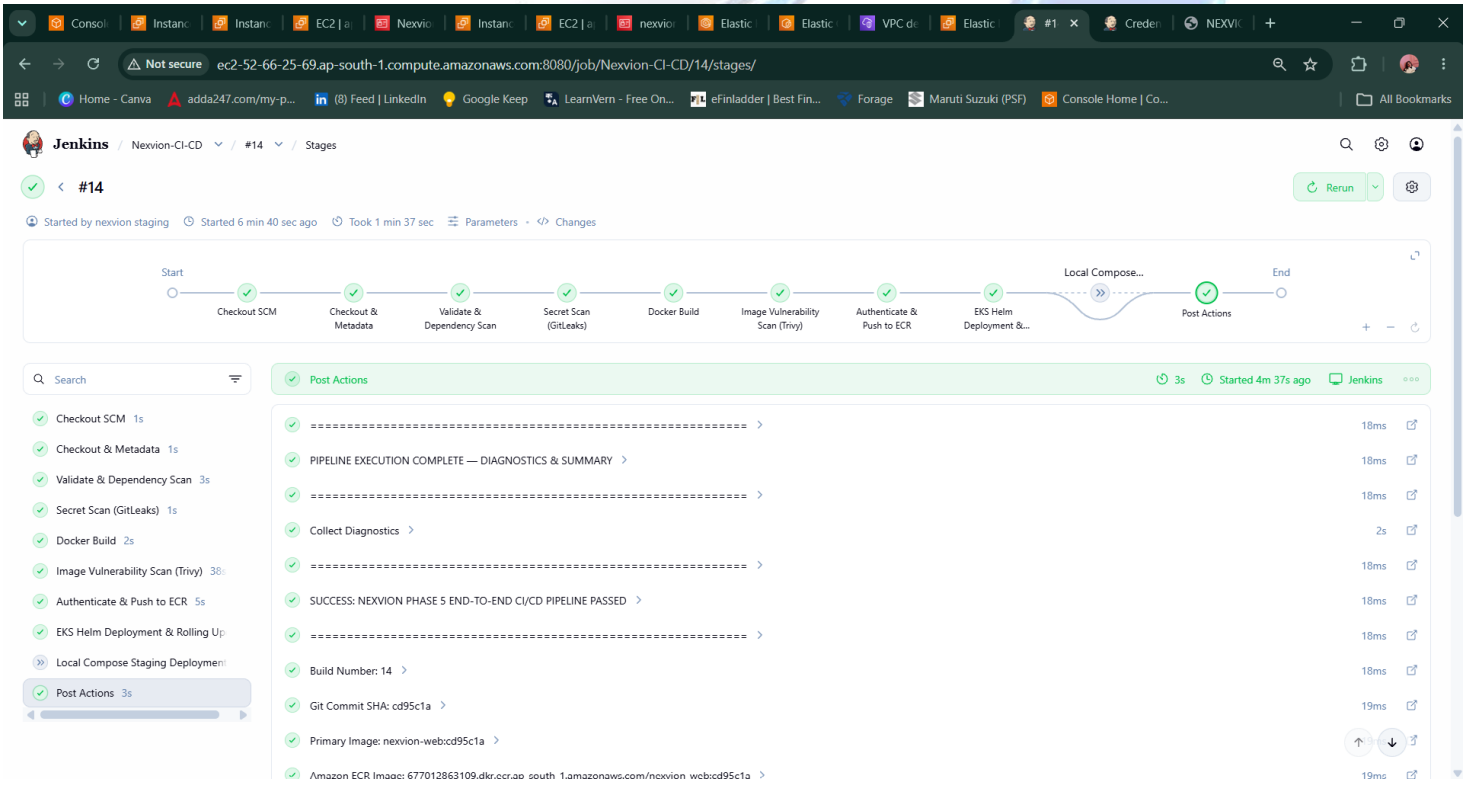
12. Project Screenshots & Evidence

12.1 Implementation Evidence

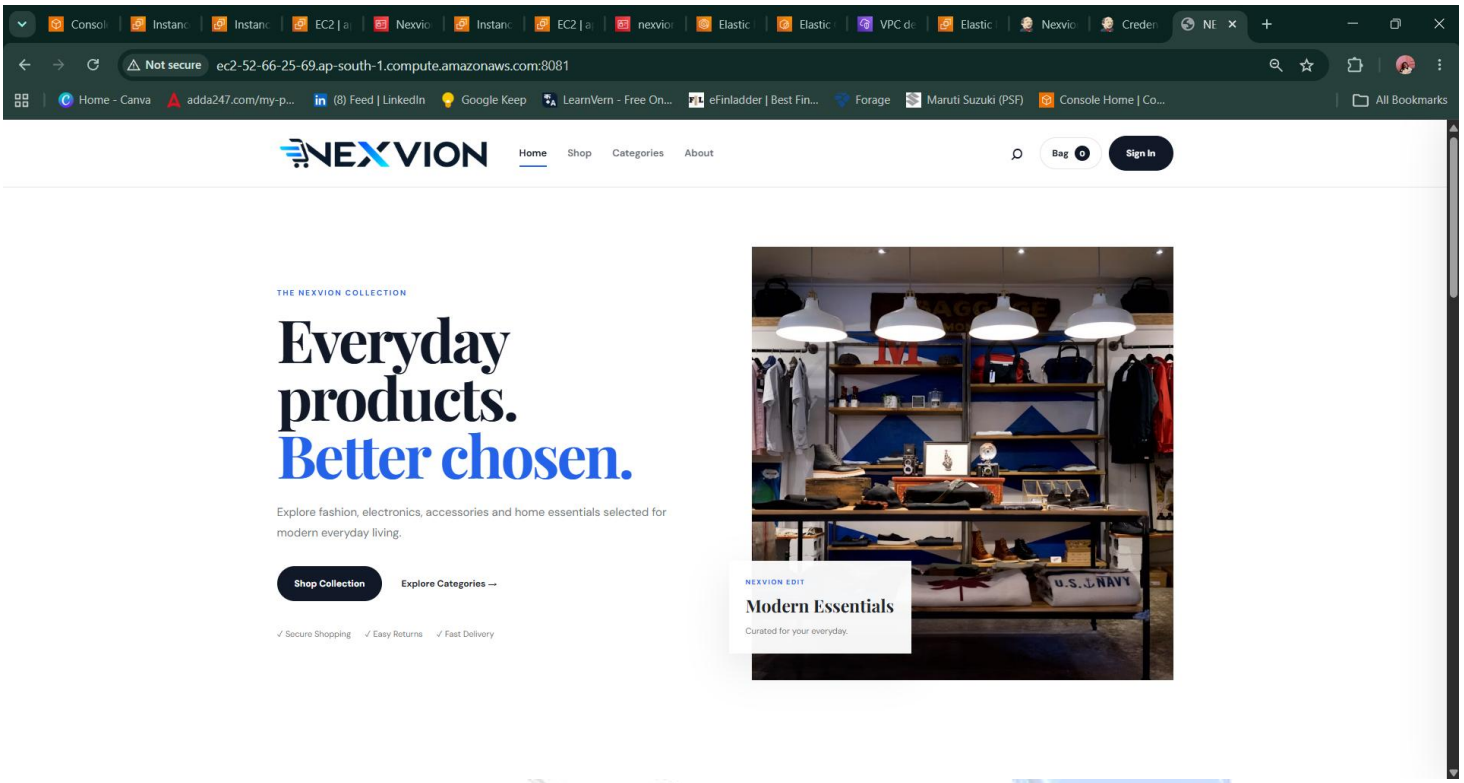
GitHub Repository URL: https://github.com/lalitpunjabi/Nexvion_DT_Project.git

All screenshots of the project are there in the “Screenshots” directory of this repository.





12.2 Final Output Evidence



13. Challenges & Solutions

During the implementation of the DevOps and cloud-native platform, several technical challenges were encountered and resolved.

Challenge	Solution
Containerization	Created a hardened Docker/NGINX setup with health checks, non-root execution and restricted filesystem access.
CI/CD Security	Integrated GitLeaks, npm audit and Trivy into Jenkins to detect secrets, dependency issues and image vulnerabilities.
Kubernetes Resource Limits	The single t3.small worker had limited pod capacity. The deployment strategy was optimized using maxSurge: 0 and maxUnavailable: 1.
AWS/EKS Deployment	Terraform was used for infrastructure provisioning, while ECR and Helm were used for image management and Kubernetes deployment.
HPA Metrics	Installed Metrics Server to provide live CPU metrics required by the Horizontal Pod Autoscaler.
Centralized Logging	Implemented Fluent Bit, Elasticsearch and Kibana with resource limits suitable for the staging environment.
Deployment Failure Recovery	Jenkins was configured to collect diagnostics, analyze incidents and dynamically rollback to the previous successful Helm revision.
AI Incident Analysis	Developed a Python-based analysis tool integrating Kubernetes, Elasticsearch and Prometheus data with Gemini, while keeping API credentials outside Git.

13.1 Implementation Evidence

The challenges were resolved through automation, infrastructure-as-code, security controls, Kubernetes resource optimization, observability and automated rollback. These solutions helped establish a reliable and repeatable DevOps workflow while clearly identifying the limitations of the current staging environment.

14. Learning Outcomes

Through this project, the following technical and practical skills were developed:

- **DevOps & CI/CD:** Practical experience with Jenkins, automated builds, testing, security scanning, image publishing and deployment.
- **Containerization:** Hands-on experience with Docker, Docker Compose and secure container configuration.
- **Cloud & Infrastructure:** Practical implementation of AWS services, Terraform and Ansible for infrastructure and configuration automation.
- **Kubernetes:** Experience with Amazon EKS, Helm, deployments, services, ingress, HPA, health probes and rolling updates.
- **DevSecOps:** Understanding and implementation of GitLeaks, dependency scanning and Trivy-based container security.
- **Monitoring & Logging:** Hands-on experience with Metrics Server, Prometheus, Grafana, Fluent Bit, Elasticsearch and Kibana.
- **Incident Management:** Experience in diagnosing deployment failures, analyzing logs and performing Helm-based rollback.
- **AI-Assisted Operations:** Practical understanding of using AI with Kubernetes, Elasticsearch and Prometheus data for incident analysis.
- **Problem Solving:** Improved ability to troubleshoot infrastructure, networking, deployment, resource and configuration issues.
- **End-to-End Engineering:** Developed an understanding of the complete lifecycle from **source code → CI/CD → cloud deployment → monitoring → logging → incident recovery**.

14.2 Overall Learning

The project provided practical exposure to building an integrated DevOps and cloud-native delivery environment rather than working with individual tools in isolation. It strengthened both technical troubleshooting skills and understanding of automated software delivery practices.

15. Conclusion

15.1 Project Conclusion

The project successfully established an integrated **DevOps and cloud-native delivery platform** around the existing Nexvion e-commerce application. Instead of focusing on redevelopment of the application itself, the project focused on creating a reliable ecosystem for building, securing, deploying, monitoring, and maintaining the application.

The implemented workflow connects **Git/GitHub, Jenkins, Docker, security scanning, Amazon ECR, Terraform, Ansible, Amazon EKS, Helm, Prometheus, Grafana, ELK, and AI-assisted incident analysis** into a connected delivery lifecycle. This enables the application to move from source code through validation, security checks, containerization, registry publishing, Kubernetes deployment, health verification, monitoring, logging, and incident analysis.

The final implementation was validated through successful CI/CD execution, container vulnerability scanning, ECR image publishing, EKS deployment, Kubernetes health checks, HPA metrics, Prometheus/Grafana monitoring, centralized log collection, and controlled deployment failure recovery using Helm rollback.

The project also provided practical exposure to real-world DevOps challenges such as resource constraints, deployment failures, security validation, Kubernetes scheduling, centralized logging, and infrastructure automation. The staging environment limitations, including the single-node EKS configuration and internal NodePort-based ingress, were documented clearly rather than treating the environment as a production deployment.

Overall, the project achieved its primary objective of developing a **secure, automated, observable, and repeatable software delivery platform**. It demonstrates how multiple DevOps and cloud-native technologies can be integrated into one complete lifecycle and provides a foundation that can be further extended toward a highly available production environment.

16. Future Enhancements

16.1 Possible Improvements

The current project provides a functional **staging-level DevOps and cloud-native platform**. The following enhancements can be implemented in future to move the platform toward production readiness:

- **High Availability:** Deploy multiple EKS worker nodes across multiple Availability Zones.
- **Production Ingress:** Replace the internal NodePort setup with an AWS Load Balancer and configure Route 53 and TLS/HTTPS.
- **Persistent Storage:** Use Amazon EBS or other persistent storage for Elasticsearch and monitoring components.
- **Advanced Secret Management:** Integrate AWS Secrets Manager or External Secrets Operator instead of environment-based secret configuration.
- **GitOps:** Introduce Argo CD for declarative and automated Kubernetes deployments.
- **Advanced Deployment Strategies:** Implement blue-green or canary deployments for safer production releases.
- **Enhanced Security:** Add SBOM generation, image signing, policy-as-code and additional compliance checks.
- **Scalable Observability:** Move to highly available Prometheus, Grafana and centralized logging architecture.
- **Disaster Recovery:** Implement automated backups, recovery procedures and multi-AZ disaster recovery.
- **AI Enhancement:** Extend the incident analysis system with historical incident correlation, automated recommendations and predictive analysis.

These enhancements would improve **availability, scalability, security, reliability, and operational maturity**, allowing the current staging implementation to evolve into a production-grade DevOps platform.

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17.1 Resources Used

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